

Machine Learning–Based Employee Attrition Prediction and Risk Scoring System

Bindu M

Unified Mentor – Data Analysis Internship

bindugehlothm@gmail.com

Abstract - Employee attrition continues to be a significant and costly challenge for organizations, and Palo Alto Networks currently does not have a systematic quantitative approach for identifying employees who may be at risk of leaving. To address this gap, this paper presents a machine learning-based system developed using a dataset of 1,470 employee records with an attrition rate of 16%. Exploratory analysis identified overtime, job satisfaction, and work-life balance as important behavioral indicators of attrition. In addition, four engineered features were introduced to strengthen the predictive capability of the models. Three classification algorithms — Logistic Regression, Random Forest, and XGBoost — were trained and evaluated using five performance metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Among the three models, XGBoost was selected as the final model because it achieved the strongest overall balance of performance, with an F1-Score of 0.459, Precision of 0.630, and ROC-AUC of 0.810. Although Random Forest produced the highest raw Accuracy at 0.840, this result was misleading because it was accompanied by a much lower Recall of 0.213, indicating limited ability to correctly identify actual attrition cases. The final model generates a probability-based attrition score for each employee and classifies employees into Low, Medium, and High Risk categories, with 210 employees, or 14.3%, placed in the High Risk group. To improve interpretability, SHAP-based explainability is also used to provide individualized reason codes that highlight the key factors influencing each employee's risk score. These outputs are presented through a deployed Streamlit dashboard, allowing Human Resources teams to shift from reactive retention efforts to proactive, data-driven workforce management.

Index Terms – Employee Attrition, Explainable AI, Risk Scoring, XGBoost.

I. INTRODUCTION

Employee attrition is one of the more difficult problems in workforce management, largely because it tends to be diagnosed only after the fact. By the time an employee formally resigns, most of the decision-making that led to that outcome has already taken place, leaving organizations to respond through counter-offers or rushed replacement hiring rather than through any form of prevention. Palo Alto

Networks faces this same difficulty: sudden resignations, the loss of high-performing employees, and reactive retention efforts that arrive too late to be effective. At present, the organization has no systematic way to anticipate which employees are likely to leave, no quantitative measure of individual attrition risk, and limited visibility into the specific factors driving these decisions.

This paper addresses a single central question: *which specific employees are most likely to leave in the near future, and why?* Answering this question requires more than a simple binary prediction. A useful system must also express *how confident* that prediction is, in a form that can be prioritized and acted upon, and it must be able to explain its reasoning in terms that Human Resources personnel can trust and interpret, rather than functioning as an opaque black box.

To this end, the present work develops a complete attrition prediction and risk scoring pipeline using a dataset of 1,470 employee records. The approach proceeds in four stages. First, exploratory data analysis is used to identify behavioural and demographic patterns associated with attrition. Second, three classification models of increasing complexity are trained and evaluated, with particular attention paid to the risk of misleading accuracy scores under class imbalance. Third, the selected model is used to generate a continuous attrition probability for every employee, which is then translated into Low, Medium, and High Risk categories to support prioritization. Finally, SHAP-based explainability is applied at both the global and individual level, allowing the system to describe not only *who* is at risk, but the specific combination of factors contributing to that individual's classification.

The resulting system is deployed as an interactive Streamlit dashboard, allowing these outputs to be explored directly rather than delivered solely as a static report.

II. DATASET DESCRIPTION

The dataset used in this study consists of 1,470 employee records, each described by 31 attributes spanning demographic information, compensation, job characteristics, and self-reported satisfaction measures. The target variable, Attrition, indicates whether an employee left the organization (1) or remained (0). Of the 1,470 employees, 237 (16.1%) are labelled as having left, while

1,233 (83.9%) remained, reflecting a class imbalance that was accounted for during model training.

The dataset contains no missing values and no duplicate records, requiring no imputation or row-level cleaning prior to analysis. Features include both numerical attributes (e.g., Age, MonthlyIncome, TotalWorkingYears, YearsAtCompany) and categorical attributes (e.g., Department, JobRole, MaritalStatus, BusinessTravel, OverTime), the latter of which were encoded prior to modelling, as described in Section III. Several features are recorded on ordinal satisfaction scales ranging from 1 (Low) to 4 (High), including JobSatisfaction, EnvironmentSatisfaction, JobInvolvement, and WorkLifeBalance.

Table 1 summarizes the key characteristics of the dataset.

Table 1. Dataset Summary

Property	Value
Total employees	1,470
Total features	31
Attrition rate	16.1% (237 employees)
Missing values	None
Duplicate records	None
Feature types	Numerical and categorical

III. METHODOLOGY

I. Data Preprocessing

Prior to model training, categorical variables were converted into numerical form using label encoding for binary attributes (OverTime, Gender) and one-hot encoding for multi-category attributes (Department, JobRole, MaritalStatus, BusinessTravel, EducationField). Numerical features were then standardized using a StandardScaler, ensuring that features on larger scales, such as MonthlyIncome, did not disproportionately influence model training relative to smaller-scale features such as satisfaction ratings.

The dataset was split into training and testing subsets using an 80/20 ratio with stratification on the target variable, preserving the original 16.1% attrition rate in both subsets. To address the resulting class imbalance in the training data, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training set, generating

synthetic minority-class samples until the training data reflected a balanced 986/986 distribution between the two classes. The test set was left unaltered to ensure that model evaluation reflected real-world class proportions.

II. Feature Engineering

Four additional features were engineered to capture relationships between existing variables that were considered likely to influence attrition but were not directly represented in the raw dataset:

- Income-to-Experience Ratio, calculated as MonthlyIncome divided by (TotalWorkingYears + 1), intended to capture whether an employee's compensation is proportionate to their level of experience.
- Promotion Delay, calculated as YearsSinceLastPromotion divided by (YearsAtCompany + 1), representing the proportion of an employee's tenure that has elapsed without promotion.
- Engagement Score, computed as the mean of JobSatisfaction, EnvironmentSatisfaction, JobInvolvement, and WorkLifeBalance, providing a single composite measure of overall employee engagement.
- Workload Stress Flag, a binary indicator set when an employee both works overtime and reports a WorkLifeBalance rating of 2 or below, identifying a combined overwork-and-dissatisfaction condition.

III. Model Development

Three classification models were trained on the preprocessed and balanced training data. Logistic Regression was employed as an interpretable baseline model. Random Forest, an ensemble method combining the predictions of 200 independently trained decision trees, was employed as a more expressive alternative. XGBoost, a gradient boosting algorithm that iteratively constructs trees to correct the errors of preceding trees, was employed as the most advanced of the three models. All models were trained using a fixed random seed to ensure reproducibility, and evaluated on the untouched, class-imbalanced test set to reflect real-world deployment conditions.

IV. EXPLORATORY DATA ANALYSIS AND INSIGHTS

A. Overtime

Employees who work overtime were found to leave at a substantially higher rate than those who do not. Approximately 30% of overtime employees left the organization, compared to approximately 12% of non-overtime employees — a rate difference of roughly 2.5 times. This pattern proved consistent throughout the analysis, and overtime status was later confirmed as the single strongest predictor of attrition in the trained model (Section VII).

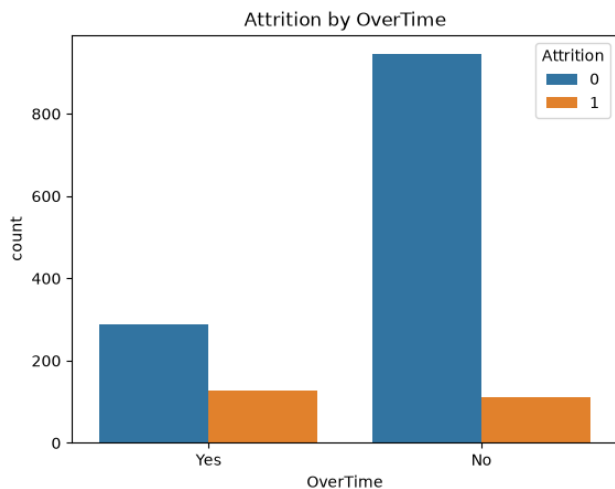


Figure 1. Attrition by overtime status.

B. Job Satisfaction and Work-Life Balance

A clear inverse relationship was observed between job satisfaction and attrition: employees reporting the lowest satisfaction level (rating 1) left at approximately 24%, compared to approximately 16% among those reporting the highest level (rating 4). A similar, more pronounced pattern was observed for work-life balance, where employees reporting the poorest rating (1) exhibited the highest attrition rate of any group examined in this study, at approximately 33%.

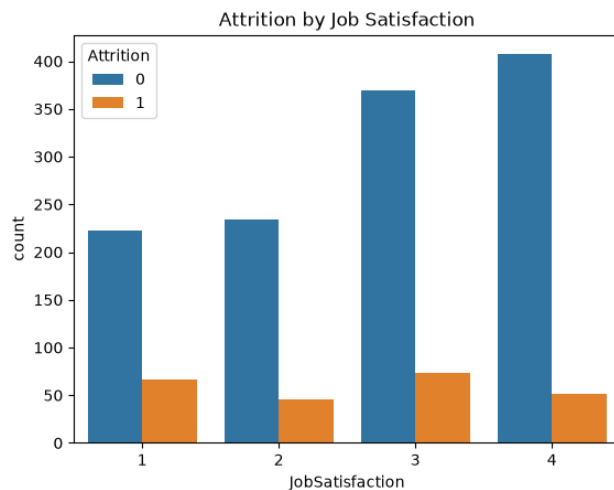


Figure 2. Attrition by job satisfaction rating

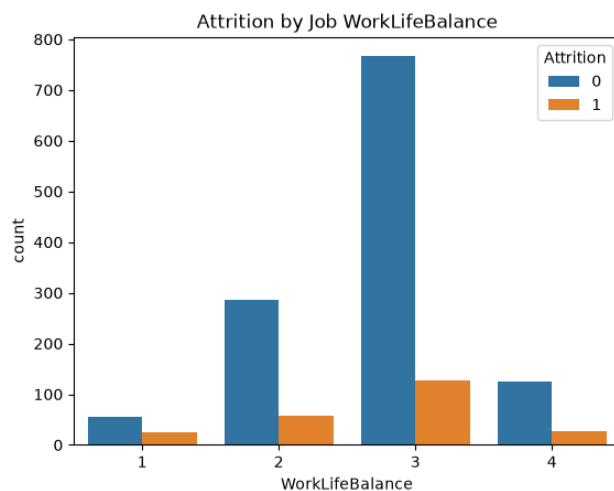


Figure 3. Attrition by work-life balance rating

C. Department and Job Role

Attrition rates were found to vary meaningfully by department and job role, with certain roles exhibiting disproportionately higher rates of departure relative to others. This variation is discussed further in Section VI in the context of department-level risk aggregation.

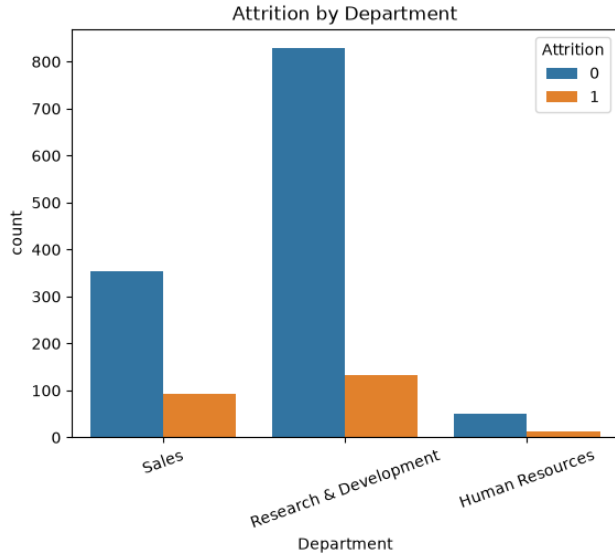


Figure 4. Attrition by department

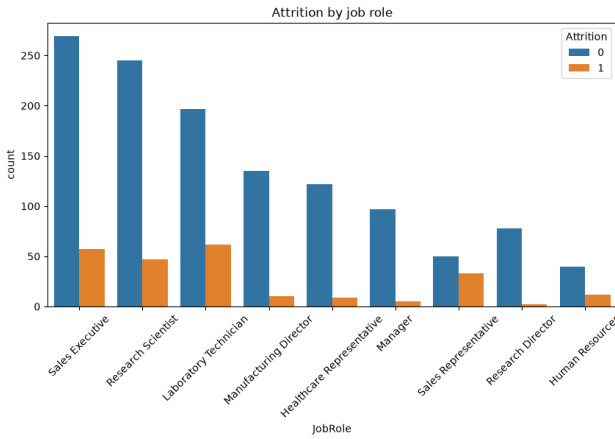


Figure 5. Attrition by job role

D. Correlation Analysis

A correlation analysis of numerical features against Attrition revealed no single variable with a strong linear relationship to the target; the strongest correlation observed was TotalWorkingYears ($r = -0.171$), followed by JobLevel ($r = -0.169$), YearsInCurrentRole ($r = -0.161$), and MonthlyIncome ($r = -0.160$). The negative direction of these correlations indicates that more experienced, senior, and higher-compensated employees were, on average, less likely to leave. The absence of any strongly correlated single feature suggests that attrition in this dataset is driven by a combination of interacting factors rather than any one dominant variable, motivating the use of ensemble and boosting-based models capable of capturing such interactions.

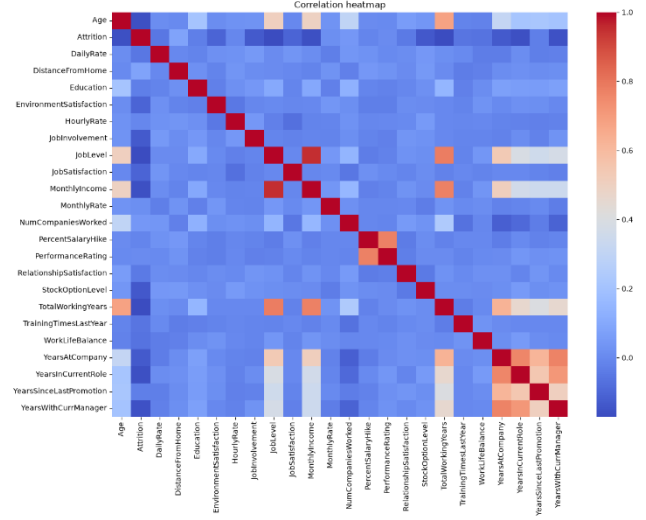


Figure 6. Correlation heatmap of numerical features

V. MODEL EVALUATION AND RESULTS

Table 2 presents the performance of the three trained models, evaluated on the untouched, class-imbalanced test set (247 employees who remained, 47 who left).

Table 2. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.772	0.365	0.574	0.446	0.791
Random Forest	0.840	0.500	0.213	0.299	0.793
XGBoost	0.864	0.630	0.362	0.459	0.810

Although Random Forest achieved the second-highest accuracy, its confusion matrix showed a Recall of only 0.213, meaning it correctly identified just 10 of the 47 actual attrition cases in the test set. This highlights a well-known limitation of using accuracy alone in imbalanced classification problems, since a model may appear to perform well numerically while still failing to detect a large share of the positive cases that matter most in practice.

Table 3 presents the confusion matrix for each model, where each row indicates the number of employees the model correctly or incorrectly classified as having stayed or left.

Table 3. Confusion Matrices

Model	Predicted Stay (Actual Stay)	Predicted Leave (Actual Stay)	Predicted Stay (Actual Leave)	Predicted Leave (Actual Leave)
Logistic Regression	200	47	20	27
Random Forest	237	10	37	10
XGBoost	237	10	30	17

Logistic Regression achieved the highest Recall among the three models, correctly identifying 27 of the 47 attrition cases, although this came with a noticeably higher false-positive rate. XGBoost, on the other hand, delivered the best overall balance of performance, achieving the highest F1-Score of 0.459, Precision of 0.630, and ROC-AUC of 0.810. For this reason, XGBoost was chosen as the final model for risk scoring and explainability analysis in the following sections.

This selection reflects a deliberate emphasis on balanced classification performance and ranking quality, particularly ROC-AUC, which is especially relevant for the probability-based risk scoring approach used later in the paper, rather than relying solely on the slightly higher Recall achieved by Logistic Regression.

VI. RISK SCORING FRAMEWORK

A model that only outputs "will leave" or "will stay" is of limited practical use to HR, since it gives no sense of how confident that prediction is or which cases deserve immediate attention. To address this, the trained XGBoost model was used to generate a continuous attrition probability for every employee in the dataset, using its `predict_proba()` output rather than a fixed 0/1 classification.

Each probability was then mapped to one of three risk categories, following the thresholds specified in the project brief:

- Low Risk: probability below 30%
- Medium Risk: probability between 30% and 60%
- High Risk: probability above 60%

Applying this framework to the full dataset of 1,470 employees produced the distribution shown in Table 4.

Table 4. Risk Category Distribution

Risk Category	Employees	Percentage
Low Risk	1,243	84.6%
Medium Risk	17	1.2%
High Risk	210	14.3%

The relatively small Medium Risk group is notable. Rather than distributing probabilities smoothly across the 0-100% range, the model tends to push most predictions toward the extremes — either confidently low or confidently high. This is a known characteristic of tree-based ensemble models like XGBoost, and it is not a flaw in the categorization thresholds themselves.

It is also worth noting that the 210 employees flagged High Risk is reasonably close to the 237 employees who actually left in the historical data, which offers some reassurance that the model's risk categories are, at least at a high level, tracking something real rather than producing an arbitrary split.

This risk table, comprising each employee's department, job role, attrition probability, and risk category, was exported and used directly to power the Streamlit dashboard described in later part of the paper.

VII. EXPLAINABILITY

A risk score alone does not tell HR *why* an employee has been flagged, which limits the practical trust and actionability of the system. This section addresses that gap at two levels: global explainability, which identifies the factors driving attrition across the organization as a whole, and individual explainability, which identifies the specific factors behind a single employee's risk classification.

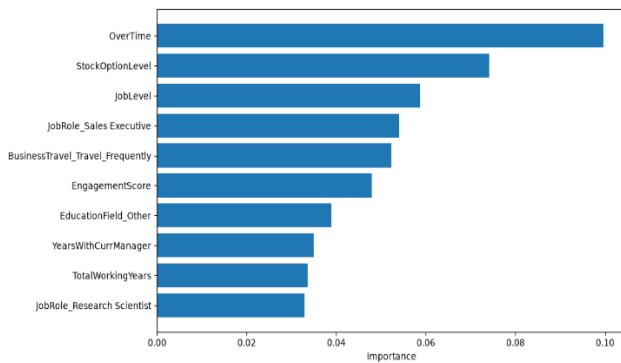
A. Global Feature Importance

Feature importance scores were extracted directly from the trained XGBoost model, reflecting how frequently and how effectively each feature was used across its decision trees. Table 5 lists the ten most important features identified.

Table 5. Top 10 Feature Importances

Rank	Feature	Importance
1	OverTime	0.099
2	StockOptionLevel	0.074
3	JobLevel	0.059
4	JobRole_Sales Executive	0.054
5	BusinessTravel_Travel_Frequently	0.052
6	EngagementScore	0.048
7	EducationField_Other	0.039
8	YearsWithCurrManager	0.035
9	TotalWorkingYears	0.034
10	JobRole_Research Scientist	0.033

OverTime emerged as the most influential feature by a clear margin, consistent with the disproportionately higher attrition rate observed for overtime employees in Section IV. Notably, the engineered Engagement Score feature (Section III) ranked sixth among 48 total features, suggesting that combining four individual satisfaction-related variables into a single composite measure provided genuine additional predictive value beyond what the original variables offered individually. The presence of BusinessTravel_Travel_Frequently among the top features further indicates that frequent travel is associated with elevated attrition risk, a relationship not directly examined during the exploratory analysis in Section IV.

**Figure 7. Top 10 feature importances (XGBoost).**

B. Individual Explainability with SHAP

While feature importance identifies patterns at the population level, it does not explain why any single employee received their particular risk score. To address this, SHAP (SHapley Additive exPlanations) was applied,

providing a fair, mathematically grounded attribution of each feature's contribution to an individual prediction.

Applying SHAP to three separate employees in the test set produced three distinct sets of reason codes:

- Employee A: MonthlyIncome, YearsWithCurrManager, Age
- Employee B: OverTime, NumCompaniesWorked, StockOptionLevel
- Employee C: StockOptionLevel, YearsSinceLastPromotion, DistanceFromHome

These results confirm that the explanations generated are genuinely individualized rather than a generic restatement of the global feature importance ranking; each employee's flagged risk stems from a distinct combination of factors relevant to their own circumstances. This individual-level reasoning was integrated directly into the Streamlit dashboard, allowing HR personnel to view a personalized explanation for any employee on demand, computed live using the saved SHAP explainer.

VIII. DASHBOARD OVERVIEW

The complete attrition prediction and risk-scoring system was deployed as an interactive web application, making the model outputs accessible to HR personnel without requiring direct use of code or notebooks. The dashboard is organized into five sections.

The Overview tab provides a company-wide summary of employees across the Low, Medium, and High Risk categories, along with an attrition-risk-by-tenure view. Because the dataset contains no transactional or calendar dates, company tenure was used as the closest available proxy for time-based analysis. This tab also includes a CSV export option for offline review.

The Department View tab summarizes attrition risk by department and job role, showing both average attrition probability and the number of employees classified as High Risk. This helps HR teams identify departments or roles that may require immediate attention.

The Employee Profile tab allows a user to select a specific employee by ID and view their individual attrition probability, risk category, and SHAP-based explanation of the factors contributing to that score. This supports case-level interpretation for individual employees.

The Explainability tab presents the global feature-importance ranking described earlier and includes a what-if scenario tool. This tool lets the user adjust inputs such as OverTime status or Work-Life Balance rating and observe

the resulting change in predicted attrition probability, which is useful for exploratory retention analysis.

Finally, the About This Analysis tab documents the dataset, modeling approach, and risk-threshold definitions for reference. All filters — including department, job role, risk category, and minimum probability threshold — apply consistently across every tab, ensuring a unified view of the data. The dashboard has been deployed and is publicly accessible at the provided Streamlit link – <https://attrition-risk-dashboard.streamlit.app/>

IX. CONCLUSION AND RECOMMENDATIONS

This paper presented a complete attrition prediction and risk scoring system developed for Palo Alto Networks, addressing the organization's stated need for a systematic, quantitative, and explainable approach to identifying employees at risk of leaving. Using a dataset of 1,470 employees, exploratory analysis identified overtime, low job satisfaction, and poor work-life balance as behavioral indicators strongly associated with attrition, while four engineered features were shown to add meaningful predictive value beyond the original dataset variables. Among three models evaluated, XGBoost was selected as the final model on the basis of its superior F1-Score, Precision, and ROC-AUC, a decision that specifically avoided the pitfall of selecting Random Forest on the basis of raw accuracy alone, despite its considerably weaker ability to correctly identify actual attrition cases.

The resulting system assigns each employee a probability-based risk score, categorized into Low, Medium, and High Risk tiers, with 210 employees (14.3%) currently flagged as High Risk. SHAP-based explainability further ensures that each flagged employee is accompanied by an individualized explanation of the factors contributing to their risk, rather than a generic, one-size-fits-all justification. These outputs are made available through a deployed, interactive dashboard, allowing HR personnel to explore risk at the company, department, and individual level without requiring technical expertise.

Based on these findings, the following recommendations are proposed for Palo Alto Networks:

- Overtime policy review. Given that OverTime was identified as the single strongest predictor of attrition, departments with high concentrations of overtime work should be prioritized for workload redistribution or staffing review.
- Targeted retention outreach. The 210 employees currently flagged High Risk represent a manageable, prioritized list for proactive retention conversations, rather than requiring HR to act reactively across the entire workforce.

- Department-level attention. Departments identified in Section VI as carrying disproportionately higher average risk should be examined individually to determine whether attrition drivers are role-specific or reflective of broader management or workload conditions within that department.
- Early-career support. Given the observed relationship between lower tenure, lower job level, and higher attrition risk, structured onboarding and early-career mentorship programs may help address attrition among newer employees before it becomes a pattern.

Future work could extend this system by incorporating longitudinal data, such as periodic satisfaction surveys or performance review history, to capture attrition risk as it evolves over time rather than as a single static snapshot. Additionally, integrating the what-if scenario tool with cost data — for example, estimating the financial impact of a promotion or workload adjustment against the cost of employee turnover — could further support HR decision-making by quantifying the trade-offs involved in specific retention interventions.

X. REFERENCES

- [1] IBM, "HR Analytics Employee Attrition & Performance Dataset," accessed via Unified Mentor project portal, 2026.
- [2] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [4] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, 2017.
- [5] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

XI. ABOUT THE AUTHOR

Bindu is a data analyst student completing an internship program with Unified mentor, through which she was assigned a first hands-on work with machine learning and predictive modelling, undertaken as a part of that program to build practical knowledge and experience beyond her core data analysis training.